AyuAware Supplements Section

Technical Product Requirements Document

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Table of Contents

1. Executive Summary
2. Section Overview
3. User Journey Breakdown
4. Technical Architecture
5. AI Integration & Prompting Strategy
6. Data Models & State Management
7. Component Requirements
8. API Requirements
9. Success Metrics
10. Implementation Considerations

1. Executive Summary

What We're Building

The Supplements section transforms herb and supplement guidance from a product catalog into an **intelligent Ayurvedic herbalism companion**. It answers the fundamental question: "*What herbs should I be taking, and are they actually working for me?*"

This isn't a supplement store with a health veneer. It's a personalized herbalist that:

Capability	What We Do	Why It Matters
Constitutional Matching	Recommend herbs based on Prakriti & Vikriti	Wrong herbs can aggravate doshas
Timing Optimization	Guide when and how to take each herb	Timing affects absorption and efficacy
Synergy Analysis	Show how herbs work together	Combinations can enhance or conflict
Effectiveness Tracking	Measure if herbs are actually working	Data-driven continuation decisions
Safety Monitoring	Check interactions and contraindications	Prevent harm from herb-drug or herb-herb conflicts

Why It Matters

Most supplement apps are glorified shopping carts. Users buy based on marketing claims, take herbs randomly, never track effectiveness, and have no idea if their stack makes sense together.

Ayurveda has a 5,000-year tradition of herbalism with sophisticated understanding of:

- Which herbs suit which constitutional types
- How herbs work synergistically (and antagonistically)
- Proper preparation, timing, and carriers (anupanas)
- How long to take herbs and when to stop
- Safety considerations and contraindications

Key Differentiator

Typical supplement app: "Ashwagandha is good for stress. Buy now."

AyuAware Supplements: "Ashwagandha is perfect for your elevated Vata and stress patterns. As a Pitta-dominant person, monitor for increased heat. Take 500mg morning with warm water. It synergizes well with your Triphala. After 42 days, your HRV has improved 24% and sleep quality is up 6%. Continue for 90 days total for deep nourishment."

Education first, commerce second.

2. Section Overview

2.1 Information Architecture

```
Supplements Section
│
├── 💊 Today's Supplement Schedule
│    ├── What to take today
│    ├── Timing windows (morning/afternoon/evening)
│    ├── Progress tracking
│    ├── Quick logging
│    └── Reminders
│
├── 🌿 Your Current Stack
│    ├── Active supplements list
│    ├── Why you're taking each
│    ├── Duration tracking
│    ├── Effectiveness ratings
│    └── Individual herb details
│
├── 💡 Recommendations for You
│    ├── Based on Prakriti (constitution)
│    ├── Based on Vikriti (imbalances)
│    ├── Based on health goals
│    ├── Based on symptoms
│    └── Seasonal considerations
│
```

```
├── 🤝 Synergy Analysis
│   ├── How herbs work together
│   ├── Complementary actions
│   ├── Potential conflicts
│   └── Overall synergy score
│
├── 📈 Effectiveness Tracking
│   ├── Before/after metrics
│   ├── Symptom changes
│   ├── Subjective feedback
│   └── Continue/discontinue recommendations
│
├── ⚕ Safety & Interactions
│   ├── Herb-herb interactions
│   ├── Drug interactions
│   ├── Contraindications
│   └── Monitoring alerts
│
├── 📚 Herb Library
│   ├── Ayurvedic herbs database
│   ├── Properties and uses
│   ├── Dosage guidelines
│   └── Contraindications
│
└── 🛒 Shop Integration
    ├── Quality-vetted products
    ├── Constitutional matching
    └── Trusted brands
```

2.2 The Ayurvedic Herbalism Philosophy

Herbs as Intelligent Medicine

In Ayurveda, herbs aren't just chemical compounds — they have intelligence (Prabhava) that works with the body's intelligence. Key principles:

Constitutional Matching:

Herb Type	Best For	Caution For
Warming herbs (Ashwagandha, Ginger)	Vata, Kapha	Pitta
Cooling herbs (Brahmi, Shatavari)	Pitta	Kapha
Stimulating herbs (Tulsi, Trikatu)	Kapha	Pitta
Grounding herbs (Ashwagandha, Bala)	Vata	Kapha

The Importance of Timing (Aushadha Sevana Kala):

Time	Best For	Why
Empty stomach (Abhakta)	Rasayanas, adaptogens	Maximum absorption
Before meals (Pragbhakta)	Digestive stimulants	Prepares Agni
With meals (Saabhakta)	Nutrients, minerals	Food aids absorption
After meals (Adhobhakta)	Digestive aids	Supports digestion
Between meals (Antarabhakta)	Specific treatments	Therapeutic timing
At bedtime (Nishi)	Nervines, laxatives	Night-time action

Anupanas (Carriers): Different carriers enhance different effects:

- Warm water: General use, increases absorption
- Milk: Nourishing, builds Ojas
- Ghee: Deep tissue penetration, brain
- Honey: Kapha-reducing, quick action
- Aloe vera juice: Pitta-cooling

2.3 Design Philosophy

Education-First Approach Every herb recommendation comes with a "why" — connecting it to the user's constitution, current state, and health goals.

Safety as Priority Interactions and contraindications are checked automatically. Warnings are prominent but not alarmist.

Effectiveness-Driven We track whether herbs are working, not just whether they're being taken. Data informs continue/discontinue decisions.

Quality Guidance Not all supplements are equal. We educate on what to look for and recommend quality-vetted products.

Protocol-Based Thinking Herbs are part of healing protocols, not isolated interventions. They work alongside nutrition, lifestyle, and other pillars.

3. User Journey Breakdown

3.1 First-Time Visitor Experience

Context: User navigates to Supplements section for the first time.

Twin Message:

"Welcome to Supplements! In Ayurveda, we use herbs to support the body's natural healing. Let me guide you..."

Educational Card:

```
💡 Ayurvedic Herbs & Supplements
─────────────────────────────────────────

In Ayurveda, herbs are used to:

✓ Balance your doshas
✓ Strengthen Agni (digestive fire)
✓ Build Ojas (vitality & immunity)
✓ Support specific organs/systems
✓ Facilitate detoxification

Unlike Western supplements focused on
individual nutrients, Ayurvedic herbs work
holistically to restore balance.

Key Principles:

🌿 Constitutional Match
Herbs should suit YOUR body type

⏰ Proper Timing
When and how you take herbs matters

🤝 Synergy
Herbs work better in combination

⚖️ Quality Matters
Source, purity, and preparation count

I'll recommend herbs specifically for your
constitution and current needs.

              [Got it! →]
```

After "Got it!", section loads with recommendations or empty state for adding first supplement.

3.2 Returning User Experience

Context: User returns to Supplements section with active stack.

Twin Message:

"Let's review your supplements and see how they're working for you..."

Immediately shows Today's Schedule card.

3.3 Viewing Today's Schedule

Context: User views daily supplement schedule.

Collapsed Today's Schedule Card:

```
┌──────────────────────────────────────────────────┐
│ 💊 Today's Supplement Schedule                   │
│ ────────────────────────────────────────────────│
│                                                  │
│ Saturday, January 31                             │
│                                                  │
│ ☀️ Morning (7 AM)                                │
│ ✓ Ashwagandha - 500mg                            │
│ ○ Triphala - 1 capsule                           │
│                                                  │
│ 🌙 Evening (9 PM)                                │
│ ○ Triphala - 1 capsule                           │
│                                                  │
│ Progress: 1 of 3 taken today                     │
│                                                  │
│ ⏰ Next: Triphala in 30 minutes                  │
│                                                  │
│            [See Full Schedule ↓]                 │
└──────────────────────────────────────────────────┘
```

Status Indicators:

- ✓ Taken (green checkmark)
- ○ Not yet taken (empty circle)
- ⏰ Upcoming within 1 hour (clock icon)

Expanded Today's Schedule Card:

1. **Date Header**
   - "Saturday, January 31"
2. **Time Windows** (grouped)
   - Morning - Upon Waking (6-8 AM)
   - After Breakfast (8-9 AM)
   - Before Bed (9-10 PM)
3. **Each Supplement Shows:**
   - Name and dose
   - Status (taken/not taken)
   - Time taken (if logged)

    - Purpose (brief)
    - How to take (with what)
    - Action buttons: [Log now] [Skip today] [Remind me]
4. **Daily Progress**
    - "1 of 3 taken (33%)"
    - Weekly adherence: "85% (good!)"
5. **Daily Tip**
    - Contextual guidance (e.g., "Take Triphala at least 2 hours after dinner")
6. **Action Buttons**
    - [View All Supplements →]
    - [Edit Schedule →]

3.4 Viewing Your Current Stack

Context: User wants to see all active supplements.

Twin Message:

"Here's what you're currently taking and why each herb was chosen for you..."

Collapsed Stack Card:

```
┌──────────────────────────────────────────────────────────┐
│ 🌿 Your Current Supplement Stack                          │
│ ──────────────────────────────────────────────────────────│
│                                                           │
│ Active Supplements: 3                                     │
│                                                           │
│  ┌──────────────────────────────────────────────────────┐ │
│  │ [🌿] Ashwagandha                                      │ │
│  │ For: Stress & Vata balance                            │ │
│  │ Taking: 42 days                                       │ │
│  │ Effectiveness: ⭐⭐⭐⭐ (Good)                          │ │
│  └──────────────────────────────────────────────────────┘ │
│                                                           │
│  ┌──────────────────────────────────────────────────────┐ │
│  │ [🌿] Triphala                                         │ │
│  │ For: Digestion & detox                                │ │
│  │ Taking: 28 days                                       │ │
│  │ Effectiveness: ⭐⭐⭐⭐⭐ (Excellent)                    │ │
│  └──────────────────────────────────────────────────────┘ │
│                                                           │
│  ┌──────────────────────────────────────────────────────┐ │
│  │ [🌿] Turmeric                                         │ │
```

```
│ │ For: Inflammation & Ojas                    │ │
│ │ Taking: 14 days                             │ │
│ │ Effectiveness: ⭐⭐⭐ (Too early)            │ │
│ └─────────────────────────────────────────────┘ │
│                                                 │
│            [See Details for Each ↓]             │
└─────────────────────────────────────────────────┘
```

Tapping Individual Supplement → Detailed View

3.5 Viewing Supplement Detail

Context: User taps on Ashwagandha to see full details.

Twin Message:

"Let me tell you about Ashwagandha and why it's good for you..."

Expanded Supplement Detail Card Contains:

1. **Header**
    - Herb name (English and Latin)
    - Sanskrit name with meaning
    - High-quality image
    - Family and part used
2. **Why You're Taking This**
    - Date recommended
    - Original reason
    - Specific needs it addresses:
        - "✓ Your Vata is elevated (40%)"
        - "✓ Stress patterns detected"
        - "✓ Sleep quality could improve"
        - "✓ Building Ojas (vitality)"
    - Personalized explanation
3. **Ayurvedic Properties**
    - Rasa (Taste): Bitter, Astringent, Sweet
    - Virya (Energy): Heating
    - Vipaka (Post-digestive effect): Sweet
    - Guna (Qualities): Heavy, Oily
    - Dosha effects with explanations
    - Best for / Caution for
4. **Traditional Uses**
    - Sanskrit action terms (Rasayana, Balya, Medhya, etc.)
    - Modern research findings
5. **Your Dosage & Timing**
    - Amount: 500mg

    - Frequency: Once daily, morning
    - Duration: 42 days (recommend 90 days total)
    - Best time with explanation
    - What to take with (anupana)
    - What to avoid
    - How long to take
6. **Effectiveness Tracking**
    - Days taking
    - Adherence percentage
    - Chart showing metrics over time
    - Improvements observed (with data)
    - Overall rating and recommendation
7. **Safety Information**
    - General safety status
    - Interactions (drug, herb)
    - Contraindications
    - Possible side effects
    - Your personal safety status
    - Monitoring notes
8. **Quality Guidance**
    - What to look for (certifications, extraction, etc.)
    - Trusted brands
9. **Action Buttons**
    - [Log Dose Now]
    - [Edit Dosage/Timing]
    - [View Effectiveness Details →]
    - [Remove from Stack]
    - [Shop This Supplement →]

3.6 Getting Personalized Recommendations

Context: User wants to know what other herbs might help.

Twin Message:

"Based on your constitution and current state, here are supplements that could help you..."

Recommendations Card:

```
┌──────────────────────────────────────────────────────┐
│ 💡 Recommended for You                                │
│ ──────────────────────────────────────────────────── │
│                                                      │
│ Based on:
│ • Your Prakriti: Pitta 60%, Vata 25%
│ • Your Vikriti: Vata elevated by 15%
│ • Your Agni: 68% (moderate)
```

```
• Your goals: Better sleep, less stress
• Season: Winter (Vata season)

Priority Recommendations:
──────────────────────────────────────────

  ┌──────────────────────────────────────┐
  │ 🌿 Brahmi                            │
  │ Priority: High                       │
  │ For: Mental clarity, calms Pitta     │
  │ Why: Balances your primary dosha     │
  │ [Learn more →]  [Add to stack]       │
  └──────────────────────────────────────┘

  ┌──────────────────────────────────────┐
  │ 🔥 Digestive Enzymes                 │
  │ Priority: Medium                     │
  │ For: Boosting Agni (68% → 85%+)      │
  │ Why: Support digestive strength      │
  │ [Learn more →]  [Add to stack]       │
  └──────────────────────────────────────┘

          [See All Recommendations ↓]
```

Tapping "Learn more" Shows:

- Full reasoning for recommendation
- Profile match analysis
- What the herb does
- Synergy with current stack
- Recommended dosage
- Expected timeline
- Action buttons: [Add to My Stack →] [Shop →] [Not Interested]

3.7 Understanding Stack Synergy

Context: User wants to know if their herbs work well together.

Twin Message:

"Let me show you how your supplements work together..."

Synergy Analysis Card:

```
┌──────────────────────────────────────────────────────────┐
│ 🤝 Supplement Synergy Analysis                           │
│ ──────────────────────────────────────────────────────── │
```

```
Your Stack Works Together:

[Visual: Venn diagram or connection map]

Ashwagandha + Triphala:
──────────────────────────────────────────
✓ Complementary Actions
• Ashwagandha: Builds strength
• Triphala: Cleanses toxins
Together: Build + Cleanse = Balance

Ashwagandha + Turmeric:
──────────────────────────────────────────
✓ Enhanced Anti-inflammatory
• Both reduce inflammation
• Synergistic effect
• Better than either alone

Triphala + Turmeric:
──────────────────────────────────────────
✓ Digestive Support
• Triphala: Cleanses GI tract
• Turmeric: Heals inflammation
• Together: Gut health

Overall Synergy Score: 95/100
Excellent combination!

          [See Full Analysis ↓]
```

3.8 Tracking Effectiveness

Context: User wants to know if supplements are working.

Twin Message:

"Let's see how well your supplements are working..."

Effectiveness Tracking Card:

```
📈 Supplement Effectiveness
──────────────────────────────────────────
```

```
│ Ashwagandha (42 days):                                  │
│ ──────────────────────────────────────────────────────  │
│                                                         │
│ [Before/After Chart]                                    │
│                                                         │
│ Improvements:                                           │
│ ✓ HRV: 50 → 62 (+24%) ⭐⭐⭐⭐⭐                          │
│ ✓ Sleep: 77% → 82% (+6%) ⭐⭐⭐⭐                        │
│ ✓ Stress: 7/10 → 4/10 ⭐⭐⭐⭐⭐                          │
│                                                         │
│ Overall: ⭐⭐⭐⭐ Good (continue)                         │
│                                                         │
│ Triphala (28 days):                                     │
│ ──────────────────────────────────────────────────────  │
│                                                         │
│ Improvements:                                           │
│ ✓ Digestion: Excellent ⭐⭐⭐⭐⭐                        │
│ ✓ Elimination: Regular ⭐⭐⭐⭐⭐                        │
│ ✓ Energy: Stable ⭐⭐⭐⭐                                │
│                                                         │
│ Overall: ⭐⭐⭐⭐⭐ Excellent (continue)                  │
│                                                         │
│           [See Detailed Analysis ↓]                     │
└─────────────────────────────────────────────────────────┘
```

Effectiveness Categories:

- ⭐⭐⭐⭐⭐ Excellent — Clear improvements, continue
- ⭐⭐⭐⭐ Good — Noticeable benefits, continue
- ⭐⭐⭐ Too early — Not enough data yet
- ⭐⭐ Minimal — Limited benefits, consider alternatives
- ⭐ Not effective — No benefits, discontinue

3.9 Checking Safety & Interactions

Context: User wants to verify safety of their stack.

Twin Message:

"Let me check if all your supplements are safe together..."

Safety Check Card:

```
┌─────────────────────────────────────────────────────────┐
```

```
⚕ Safety & Interaction Check
──────────────────────────────────────

Overall Status: ✓ Safe

Herb-Herb Interactions:
✓ No problematic interactions detected

Contraindications:
✓ No contraindications for your profile

⚠️ Monitoring Recommended:
• Pitta elevation (you have 60%)
• Ashwagandha can increase heat
• Watch for signs: acidity, irritability

If you're taking medications:
[Add medications to check interactions]

          [See Full Safety Info ↓]
```

Safety Status Levels:

- ✓ Safe (green): No concerns
- ⚠️ Caution (gold): Monitor recommended
- ⛔ Contraindicated (red): Do not combine

3.10 Browsing the Herb Library

Context: User wants to learn about other herbs.

Twin Message:

"Want to learn about other Ayurvedic herbs? Browse my library..."

Herb Library Card:

```
📚 Ayurvedic Herb Library
──────────────────────────────────────

Search: [🔍 Search herbs...]

Filter by:
[All] [Vata] [Pitta] [Kapha]
[Adaptogen] [Digestive] [Nervine] [Immune]
```

```
│                                                          │
│ Common Ayurvedic Herbs:                                  │
│                                                          │
│ • Ashwagandha — Adaptogen, Vata-balancing                │
│ • Brahmi — Brain tonic, Pitta-cooling                    │
│ • Triphala — Digestive, Tri-doshic                       │
│ • Turmeric — Anti-inflammatory, all doshas               │
│ • Tulsi — Adaptogen, Kapha-reducing                      │
│ • Shatavari — Rejuvenative, Pitta-cooling                │
│ • Guduchi — Immune support, Tri-doshic                   │
│ • Trikatu — Digestive fire, Kapha-reducing               │
│ • Amalaki — Rejuvenative, Pitta-cooling                  │
│ • Haritaki — Detox, Vata-balancing                       │
│                                                          │
│ [View Full Library →]                                    │
└──────────────────────────────────────────────────────────┘
```

3.11 Adding a Supplement to Stack

Context: User decides to add Brahmi to their stack.

Flow:

1. User taps "Add to stack" on recommendation
2. Twin confirms: "Great choice! Let me set up Brahmi for you..."
3. **Configuration Modal:**

```
┌──────────────────────────────────────────────────────────┐
│ Add Brahmi to Your Stack                                 │
│──────────────────────────────────────────────────────────│
│ Recommended Dosage:                                      │
│ [500mg ▼] (recommended for you)                          │
│ Frequency:                                               │
│ [Twice daily ▼]                                          │
│ Timing:                                                  │
│ ☑ Morning (with breakfast)                               │
│ ☑ Evening (before bed)                                   │
│ Set reminders?                                           │
│ [Yes ▼] at [8:00 AM] and [9:00 PM]                       │
│ Starting:                                                │
│ ○ Today  ● Tomorrow                                      │
│ [Add to My Stack →]                                      │
└──────────────────────────────────────────────────────────┘
```

4.
5. Confirmation: "Brahmi has been added! I'll remind you tomorrow morning."
6. Schedule and stack update

4. Technical Architecture

4.1 High-Level Architecture

```
┌──────────────
```

```
│   │ • Adherence        │   │ • Synergy data    │   │ • Messages
│   └────────────────────┘   └───────────────────┘
└──────────────────────────┘   │
└──────────────────────────────────────────────────────────────────────────────
──────────┘
                                                  │
                                                  ▼
┌──────────────────────────────────────────────────────────────────────────────
──────────┐
│                                  SERVICE LAYER
│
│   ┌────────────────────────┐                                   │
┌──────────────────────────────────────┐    │
│   │  Supplements Agent     │   │     Analysis Services
│   │
│   │  (Claude API)          │   │
│   │
│   │                        │   │  • Recommendation engine
│   │
│   │  • Recommendations     │   │  • Synergy calculator
│   │
│   │  • Herb education      │   │  • Effectiveness analyzer
│   │
│   │  • Safety analysis     │   │  • Safety/interaction checker
│   │
│   │  • Personalization     │   │  • Constitutional matcher
│   │
│   └────────────────────────┘
└──────────────────────────────────────┘    │
│
│
│   ┌────────────────────────┐                                   │
┌──────────────────────────────────────┐    │
│   │  Ayurvedic Herb        │   │     Notification Service
│   │
│   │  Database              │   │
│   │
│   │                        │   │  • Dose reminders
│   │
│   │  • Herb properties     │   │  • Refill alerts
│   │
│   │  • Interactions        │   │  • Check-in prompts
│   │
```

```
│   │  • Contraindications │   │
│   │                      │   │
│   └──────────────────────┘   │
└──────────────────────────────┘
```

```
                                  │
                                  ▼
┌──────────────────────────────────────────────────────────────────────────
──────────┐
│                              DATA LAYER
│
│   ┌──────────────────────┐                    │
┌──────────────────────────────┐
│   │   Local Cache        │   │      Backend APIs
│   │                      │   │
│   │                      │   │
│   │  • Schedule          │   │  • Supplement service
│   │                      │   │
│   │  • Stack data        │   │  • Recommendation service
│   │                      │   │
│   │  • Herb database     │   │  • Effectiveness service
│   │                      │   │
│   │  • Dose history      │   │  • Shop service
│   │                      │   │
│   └──────────────────────┘   │
└──────────────────────────────┘
```

4.2 Data Flow: Logging a Supplement Dose

```
User taps "Log now" on supplement
        │
        ▼
┌──────────────────────────┐
│ Optimistic UI update     │ ── Show checkmark immediately
└──────────────────────────┘
        │
        ▼
┌──────────────────────────┐
│ Record dose locally      │ ── { supplementId, timestamp, dose
}
└──────────────────────────┘
        │
        ▼
```

```
┌──────────────────────────┐
│ Update schedule state    │ ── Mark as taken, update progress
└──────────────────────────┘
          │
          ▼
┌──────────────────────────┐
│ Sync to backend          │ ── POST /supplements/log
└──────────────────────────┘
          │
          ▼
┌──────────────────────────┐
│ Update adherence         │ ── Recalculate weekly adherence
│ metrics                  │
└──────────────────────────┘
          │
          ▼
┌──────────────────────────┐
│ Check for milestones     │ ── "30 days of Ashwagandha!"
└──────────────────────────┘
```

4.3 Data Flow: Generating Recommendations

```
User opens Recommendations
          │
          ▼
┌──────────────────────────┐
│ Gather user context      │
│ • Prakriti               │
│ • Vikriti                │
│ • Current Agni           │
│ • Health goals           │
│ • Current stack          │
│ • Season                 │
└──────────────────────────┘
          │
          ▼
┌──────────────────────────┐
│ Query herb database      │ ── Filter by constitution match
└──────────────────────────┘
          │
          ▼
┌──────────────────────────┐
│ Check against current    │ ── Exclude already taking
│ stack                    │    Check for interactions
└──────────────────────────┘
          │
          ▼
```

```
┌──────────────────────────┐
│ Score candidates         │ ── Priority based on needs
└──────────────────────────┘
          │
          ▼
┌──────────────────────────┐
│ Claude generates         │ ── Personalized explanations
│ recommendation text      │    Why each herb suits them
└──────────────────────────┘
          │
          ▼
┌──────────────────────────┐
│ Display ranked           │
│ recommendations          │
└──────────────────────────┘
```

4.4 Data Flow: Effectiveness Analysis

```
User views Effectiveness
          │
          ▼
┌──────────────────────────┐
│ For each supplement:     │
│ Get start date           │
└──────────────────────────┘
          │
          ▼
┌──────────────────────────┐
│ Query health metrics     │ ── Before/after comparison
│ from start date          │    HRV, sleep, stress, etc.
└──────────────────────────┘
          │
          ▼
┌──────────────────────────┐
│ Get subjective logs      │ ── User-reported symptoms
│                          │    Mood, energy, etc.
└──────────────────────────┘
          │
          ▼
┌──────────────────────────┐
│ Calculate adherence      │ ── Days taken / days scheduled
└──────────────────────────┘
          │
          ▼
┌──────────────────────────┐
│ Correlate changes        │ ── Did metrics improve?
│ with supplement use      │
└──────────────────────────┘
```

```
└──────────────────────────┘
        │
        ▼
┌──────────────────────────┐
│ Generate effectiveness   │ ── Score, continue/discontinue
│ rating                   │
└──────────────────────────┘
        │
        ▼
┌──────────────────────────┐
│ Claude generates         │ ── Narrative explanation
│ analysis                 │
└──────────────────────────┘
```

4.5 Safety Check Flow

```
User views Safety Check (or adds new supplement)
        │
        ▼
┌──────────────────────────┐
│ Get all active           │
│ supplements              │
└──────────────────────────┘
        │
        ▼
┌──────────────────────────┐
│ Get user medications     │ ── If provided
│ (if any)                 │
└──────────────────────────┘
        │
        ▼
┌──────────────────────────┐
│ Query interaction        │ ── Herb-herb interactions
│ database                 │    Herb-drug interactions
└──────────────────────────┘
        │
        ▼
┌──────────────────────────┐
│ Check contraindications  │ ── Against user's conditions
│ against user profile     │    pregnancy, allergies, etc.
└──────────────────────────┘
        │
        ▼
┌──────────────────────────┐
│ Check constitutional     │ ── Warming herb + high Pitta?
│ concerns                 │
└──────────────────────────┘
```

```
        │
        ▼
┌──────────────────────────┐
│ Generate safety          │ ── Overall status
│ report                   │    Specific warnings
└──────────────────────────┘
```

5. AI Integration & Prompting Strategy

5.1 The Role of Claude in Supplements Section

Claude serves as a knowledgeable Ayurvedic herbalist:

Capability	Description
Recommendation Generation	Suggest herbs based on constitution and needs
Herb Education	Explain properties, uses, and benefits
Personalization	Connect herbs to specific user context
Synergy Analysis	Explain how herbs work together
Safety Guidance	Warn about interactions and contraindications
Effectiveness Interpretation	Analyze if supplements are working
Protocol Design	Create supplement protocols

5.2 System Prompt for Supplements Section

```
You are the Ayurvedic Digital Twin for AyuAware, serving as a
knowledgeable herbalist guide. Your role is to help users
understand
Ayurvedic herbs, make personalized recommendations, and
ensure safe,
effective supplementation.

## YOUR ROLE IN THIS SECTION

You are an Ayurvedic herbalist. You recommend herbs based on
constitution,
explain their properties and uses, ensure safety, and track
effectiveness.
Education comes before commerce — you teach users about
herbs, not just
sell them.

## AYURVEDIC HERBALISM PRINCIPLES
```

Herbs have multiple properties:
- Rasa (Taste): Sweet, Sour, Salty, Pungent, Bitter, Astringent
- Virya (Energy): Heating or Cooling
- Vipaka (Post-digestive effect): Sweet, Sour, or Pungent
- Guna (Qualities): Heavy/Light, Oily/Dry, etc.
- Prabhava (Special action): Unique effects

Herb Categories:
- Rasayanas (Rejuvenatives): Build Ojas, longevity
- Adaptogens: Stress resilience
- Nervines: Calm nervous system
- Digestives: Support Agni
- Alteratives: Purify blood, detox
- Carminatives: Reduce gas, bloating
- Diuretics: Support kidneys
- Immunomodulators: Boost immunity

Constitutional Matching:
- Vata types need: Warming, grounding, oily herbs
- Pitta types need: Cooling, calming herbs
- Kapha types need: Warming, stimulating, drying herbs

TIMING AND ADMINISTRATION

When to take matters:
- Empty stomach (Abhakta): Maximum absorption
- Before meals (Pragbhakta): Digestive stimulants
- With meals (Saabhakta): Nutrients
- After meals (Adhobhakta): Digestive aids
- Between meals (Antarabhakta): Therapeutic
- At bedtime (Nishi): Nervines, laxatives

Carriers (Anupanas) enhance effects:
- Warm water: General, increases absorption
- Milk: Nourishing, builds Ojas
- Ghee: Deep tissue penetration
- Honey: Kapha-reducing
- Aloe vera: Pitta-cooling

SAFETY PRINCIPLES

Always check:
1. Constitutional appropriateness
2. Current imbalances (Vikriti)

3. Herb-herb interactions
4. Herb-drug interactions
5. Contraindications (pregnancy, conditions)
6. Dosage appropriateness

Warning levels:
- Safe: Generally well-tolerated
- Caution: Monitor for side effects
- Contraindicated: Do not use

EFFECTIVENESS ASSESSMENT

Consider:
- Minimum time for effect (usually 2-8 weeks)
- Objective metrics (HRV, sleep, etc.)
- Subjective reports (energy, mood)
- Adherence (are they taking it consistently?)
- Constitutional response (some people respond faster)

Ratings:
- Excellent: Clear, measurable improvements
- Good: Noticeable benefits
- Too early: Insufficient time/data
- Minimal: Limited benefits
- Not effective: No benefits observed

COMMUNICATION APPROACH

When recommending:
- Explain WHY this herb suits them
- Connect to their specific constitution
- Describe expected benefits
- Set realistic timelines
- Cover safety considerations

When educating:
- Sanskrit terms with English explanations
- Traditional context + modern research
- Practical application
- Quality guidance

WHAT NOT TO DO

- Don't recommend without constitutional context
- Don't ignore safety considerations

```
- Don't promise specific outcomes
- Don't recommend for serious medical conditions
- Don't contradict established contraindications
- Don't push products over education
```

5.3 Context Building for Supplements Section

Every Claude call includes:

```
CONTEXT PACKAGE STRUCTURE:

## User Profile
- Name: {user_name}
- Prakriti: {prakriti_percentages}
- Primary constitution: {constitution_type}
- Current Vikriti: {vikriti_percentages}
- Current imbalances: {imbalance_list}
- Current Agni: {agni_score}%
- Current Ojas: {ojas_score}%
- Health goals: {goals_list}
- Known conditions: {conditions_list}
- Medications: {medications_list} (if provided)
- Allergies: {allergies_list}

## Current Supplement Stack
- Active supplements: {supplement_list_with_details}
- Duration on each: {days_per_supplement}
- Adherence rates: {adherence_percentages}

## Health Metrics (for effectiveness)
- Relevant metrics: {metrics_with_before_after}
- Subjective reports: {symptom_logs}

## Context
- Season: {current_season}
- Weather: {weather_conditions}

## Specific Request
- What user is asking about: {query_context}
```

5.4 Prompt Templates by Scenario

Herb Recommendation:

```
Generate personalized herb recommendations for this user.

{CONTEXT_PACKAGE}
```

Generate 3-5 prioritized recommendations including:

For each herb:

1. **Herb Name** (Sanskrit and common)
   - Priority: High/Medium/Low

2. **Why This Herb Suits Them**
   - Connect to their Prakriti
   - Address their current Vikriti
   - Support their specific goals
   - Consider seasonal factors

3. **Key Benefits**
   - What it does
   - How it helps their specific situation

4. **Synergy with Current Stack**
   - How it works with what they're taking
   - Any cautions

5. **Recommended Dosage & Timing**
   - Amount
   - When to take
   - What to take with (anupana)

6. **Expected Timeline**
   - When to expect effects

Prioritize based on:
1. Constitutional match
2. Current imbalances
3. Stated goals
4. Synergy with current stack
5. Safety profile

Supplement Detail Education:

Provide comprehensive education about this herb for the user.

{CONTEXT_PACKAGE}

Herb: {herb_name}

Include:

1. **Overview**
   - Sanskrit name and meaning
   - Traditional significance
   - Modern understanding

2. **Why They're Taking This** (personalized)
   - Their specific needs it addresses
   - How it helps their constitution
   - Connection to their goals

3. **Ayurvedic Properties**
   - Rasa, Virya, Vipaka, Guna
   - Dosha effects (with explanations)
   - Traditional uses (Sanskrit terms + meaning)

4. **Modern Research**
   - Key findings
   - Mechanisms of action

5. **Dosage & Timing**
   - Their specific dosage
   - Why this timing
   - Best carriers (anupana)
   - How long to take

6. **Safety**
   - General safety profile
   - Interactions to know
   - Contraindications
   - Side effects
   - Their personal safety status

7. **Quality Guidance**
   - What to look for
   - Forms available
   - Trusted sources

Tone: Educational, empowering, thorough

Effectiveness Analysis:

Analyze supplement effectiveness for this user.

{CONTEXT_PACKAGE}

Supplement: {supplement_name}

```
Days taking: {duration}
Adherence: {adherence_percentage}%

Before/After Data:
{metrics_comparison}

Subjective Reports:
{user_feedback}

Generate analysis including:

1. **Overall Rating** (1-5 stars)
   - Category: Excellent/Good/Too Early/Minimal/Not Effective

2. **Improvements Observed**
   - List each metric that improved
   - Include percentage/absolute change
   - Rate each (1-5 stars)

3. **Analysis**
   - Interpret the changes
   - Connect to herb's expected effects
   - Consider adherence in assessment
   - Note if timeline is sufficient

4. **Recommendation**
   - Continue, adjust, or discontinue
   - If continue: how much longer
   - If adjust: what to change
   - If discontinue: what instead

5. **Next Steps**
   - Specific actions to take

Consider:
- Minimum effective duration for this herb
- User's constitution (response patterns)
- Adherence quality
- Confounding factors
```

Safety Check:

```
Perform safety analysis for this supplement stack.

{CONTEXT_PACKAGE}
```

Current Stack:
{supplement_list_with_doses}

Medications (if any):
{medications_list}

Conditions (if any):
{conditions_list}

Check and report:

1. **Overall Safety Status**
   - Safe / Caution / Contraindicated

2. **Herb-Herb Interactions**
   - List any interactions
   - Severity level
   - What to monitor

3. **Herb-Drug Interactions** (if medications provided)
   - List any interactions
   - Severity level
   - Recommendations

4. **Contraindications**
   - Any herbs contraindicated for their conditions
   - Pregnancy/nursing considerations if relevant

5. **Constitutional Concerns**
   - Warming herbs + high Pitta → monitor
   - Other constitution-specific cautions

6. **Monitoring Recommendations**
   - What to watch for
   - Signs to stop

7. **Overall Assessment**
   - Summary of safety
   - Any actions needed

Be thorough but not alarmist. Most herb combinations are safe.

Synergy Analysis:

Analyze synergy between supplements in this stack.

```
{CONTEXT_PACKAGE}

Current Stack:
{supplement_list}

Analyze:

1. **Overall Synergy Score** (0-100)
   - Excellent (90-100): Highly complementary
   - Good (70-89): Work well together
   - Neutral (50-69): No particular synergy
   - Caution (30-49): Some competing effects
   - Poor (<30): Antagonistic

2. **Pair-wise Analysis**
   For each pair of herbs:
   - How they work together
   - Complementary effects
   - Any competing effects
   - Net benefit

3. **Stack Themes**
   - What this combination achieves together
   - Balanced building + cleansing?
   - Addressing multiple needs?

4. **Gaps or Imbalances**
   - Anything missing?
   - Too much of one action?
   - Suggestions for balance

5. **Optimization Suggestions**
   - Any timing adjustments
   - Any additions that would enhance
   - Any that could be removed
```

5.5 Herb Database Structure

Claude references an Ayurvedic herb database:

```
HERB DATABASE STRUCTURE:

Herb:
  - id: string
  - commonName: string
```

```
- sanskritName: string
- sanskritMeaning: string
- latinName: string
- family: string
- partUsed: string[]

- properties:
    - rasa: string[] (tastes)
    - virya: 'heating' | 'cooling'
    - vipaka: 'sweet' | 'sour' | 'pungent'
    - guna: string[] (qualities)
    - prabhava: string (special action)

- doshaEffect:
    - vata: 'increases' | 'decreases' | 'neutral'
    - pitta: 'increases' | 'decreases' | 'neutral'
    - kapha: 'increases' | 'decreases' | 'neutral'

- categories: string[] (adaptogen, nervine, etc.)

- traditionalUses:
    - sanskritAction: string
    - meaning: string
    - explanation: string

- modernResearch: string[]

- dosage:
    - powder: { min, max, unit }
    - capsule: { min, max, unit }
    - liquid: { min, max, unit }

- timing: string[] (when to take)
- anupanas: string[] (best carriers)
- duration: { min, max, unit } (how long to take)

- interactions:
    - herbs: { herbId, severity, description }[]
    - drugs: { drugClass, severity, description }[]

- contraindications: string[]
- sideEffects: string[]
- safetyNotes: string[]

- qualityMarkers: string[]
```

```
  - trustedBrands: string[]

Example:
{
  commonName: "Ashwagandha",
  sanskritName: "Ashwagandha",
  sanskritMeaning: "Smell of horse (gives strength)",
  latinName: "Withania somnifera",
  family: "Solanaceae",
  partUsed: ["root"],
  properties: {
    rasa: ["bitter", "astringent", "sweet"],
    virya: "heating",
    vipaka: "sweet",
    guna: ["heavy", "oily"],
    prabhava: "Balya (strength-giving)"
  },
  doshaEffect: {
    vata: "decreases",
    pitta: "increases", // slight
    kapha: "decreases"
  },
  categories: ["adaptogen", "rasayana", "nervine"],
  traditionalUses: [
    { sanskritAction: "Rasayana", meaning:
"Rejuvenative", ... },
    { sanskritAction: "Balya", meaning: "Strength-
giving", ... },
    { sanskritAction: "Medhya", meaning: "Brain
tonic", ... },
    { sanskritAction: "Nidrajanana", meaning: "Sleep-
promoting", ... }
  ],
  modernResearch: [
    "Reduces cortisol by 11-32%",
    "Improves sleep quality",
    "Enhances cognitive function",
    ...
  ],
  dosage: {
    powder: { min: 3, max: 6, unit: "grams" },
    capsule: { min: 300, max: 600, unit: "mg" }
  },
  timing: ["morning", "evening"],
  anupanas: ["warm water", "warm milk", "ghee"],
```

```
  duration: { min: 6, max: 24, unit: "weeks" },
  interactions: {
    herbs: [],
    drugs: [
      { drugClass: "thyroid medications", severity:
"moderate",
        description: "May affect thyroid levels" },
      { drugClass: "sedatives", severity: "mild",
        description: "May enhance sedative effect" }
    ]
  },
  contraindications: [
    "Pregnancy (unless directed)",
    "Hyperthyroidism",
    "Acute autoimmune flare"
  ],
  sideEffects: [
    "Digestive upset (if excessive)",
    "Drowsiness",
    "Increased heat (Pitta types)"
  ],
  qualityMarkers: [
    "Organic certification",
    "Standardized extract (5% withanolides)",
    "Root only (not leaf)",
    "Third-party tested"
  ],
  trustedBrands: [
    "Banyan Botanicals",
    "Organic India",
    "Gaia Herbs"
  ]
}
```

6. Data Models & State Management

6.1 Core Data Types

Supplement (User's active supplement):

```
Supplement:
  - id: string
  - herbId: string (reference to herb database)
  - userId: string
  - name: string
```

```
  - sanskritName: string
  - dosage: Dosage
  - schedule: DoseSchedule[]
  - startDate: Date
  - recommendedDuration: number (days)
  - reasonAdded: string
  - targetImbalances: string[]
  - targetGoals: string[]
  - isActive: boolean
  - addedAt: Date
  - deactivatedAt: Date | null
  - deactivationReason: string | null

Dosage:
  - amount: number
  - unit: 'mg' | 'g' | 'ml' | 'capsule' | 'tablet' | 'tsp'
  - form: 'capsule' | 'powder' | 'liquid' | 'tablet'

DoseSchedule:
  - id: string
  - timeOfDay: 'morning' | 'afternoon' | 'evening' |
'bedtime'
  - specificTime: string (HH:mm)
  - withFood: boolean
  - anupana: string (carrier)
  - instructions: string
  - reminderEnabled: boolean
  - reminderTime: string (HH:mm)
```

DoseLog:

```
DoseLog:
  - id: string
  - supplementId: string
  - userId: string
  - scheduledTime: Date
  - actualTime: Date | null
  - status: 'taken' | 'skipped' | 'missed' | 'pending'
  - notes: string | null
  - createdAt: Date
```

DailySchedule:

```
DailySchedule:
  - date: Date
  - doses: ScheduledDose[]
  - progress: number (taken / total)
  - adherenceRate: number (0-100)
```

```
ScheduledDose:
  - supplementId: string
  - supplementName: string
  - dosage: Dosage
  - timeWindow: 'morning' | 'afternoon' | 'evening' |
'bedtime'
  - specificTime: string
  - status: 'taken' | 'pending' | 'upcoming' | 'missed'
  - takenAt: Date | null
  - instructions: string
  - anupana: string
```

SupplementStack:

```
SupplementStack:
  - userId: string
  - supplements: Supplement[]
  - synergyScore: number (0-100)
  - synergyAnalysis: SynergyAnalysis
  - safetyStatus: SafetyStatus
  - lastUpdated: Date

SynergyAnalysis:
  - overallScore: number
  - pairAnalyses: PairAnalysis[]
  - stackThemes: string[]
  - gaps: string[]
  - optimizationSuggestions: string[]

PairAnalysis:
  - herb1Id: string
  - herb2Id: string
  - synergy: 'complementary' | 'neutral' | 'competing'
  - explanation: string
  - netBenefit: string

SafetyStatus:
  - overall: 'safe' | 'caution' | 'contraindicated'
  - herbHerbInteractions: Interaction[]
  - herbDrugInteractions: Interaction[]
  - contraindications: Contraindication[]
  - constitutionalConcerns: string[]
  - monitoringRecommendations: string[]

Interaction:
```

```
  - item1: string
  - item2: string
  - severity: 'mild' | 'moderate' | 'severe'
  - description: string
  - recommendation: string

Contraindication:
  - herbId: string
  - condition: string
  - severity: 'caution' | 'avoid'
  - reason: string
```

Recommendation:

```
Recommendation:
  - id: string
  - herbId: string
  - herbName: string
  - sanskritName: string
  - priority: 'high' | 'medium' | 'low'
  - forImbalance: string | null
  - forGoal: string | null
  - reasoning: string
  - benefits: string[]
  - synergyWithStack: string
  - suggestedDosage: Dosage
  - suggestedTiming: string
  - expectedTimeline: string
  - generatedAt: Date

RecommendationSet:
  - userId: string
  - recommendations: Recommendation[]
  - basedOn: RecommendationContext
  - generatedAt: Date
  - expiresAt: Date

RecommendationContext:
  - prakriti: DoshaPercentages
  - vikriti: DoshaPercentages
  - agniScore: number
  - goals: string[]
  - currentStack: string[]
  - season: string
```

EffectivenessData:

```
SupplementEffectiveness:
```

```
  - supplementId: string
  - supplementName: string
  - daysTaking: number
  - adherenceRate: number
  - rating: EffectivenessRating
  - category: 'excellent' | 'good' | 'too_early' | 'minimal'
| 'not_effective'
  - metricChanges: MetricChange[]
  - subjectiveChanges: SubjectiveChange[]
  - analysis: string
  - recommendation: 'continue' | 'adjust' | 'discontinue'
  - recommendationReason: string
  - generatedAt: Date

EffectivenessRating:
  - overall: number (1-5)
  - confidence: 'high' | 'medium' | 'low'

MetricChange:
  - metricName: string
  - before: number
  - after: number
  - changePercent: number
  - changeAbsolute: number
  - rating: number (1-5)
  - interpretation: string

SubjectiveChange:
  - symptom: string
  - before: number (1-10)
  - after: number (1-10)
  - improvement: boolean
```

HerbLibraryEntry (from database):

```
HerbLibraryEntry:
  - id: string
  - commonName: string
  - sanskritName: string
  - sanskritMeaning: string
  - latinName: string
  - imageUrl: string
  - categories: string[]
  - doshaEffects: DoshaEffects
  - primaryUses: string[]
  - briefDescription: string
```

```
  - isInUserStack: boolean
  - isRecommended: boolean
```

6.2 State Structure

```
SupplementsState:
  # Visit Context
  - isFirstVisit: boolean
  - lastVisit: Date | null

  # Schedule
  - todaySchedule: DailySchedule | null
  - isLoadingSchedule: boolean
  - scheduleError: Error | null

  # Stack
  - stack: SupplementStack | null
  - isLoadingStack: boolean
  - activeSupplementDetail: Supplement | null

  # Recommendations
  - recommendations: RecommendationSet | null
  - isLoadingRecommendations: boolean

  # Effectiveness
  - effectivenessData: Record<string,
SupplementEffectiveness>
  - isLoadingEffectiveness: boolean

  # Safety
  - safetyStatus: SafetyStatus | null
  - isCheckingSafety: boolean

  # Herb Library
  - libraryEntries: HerbLibraryEntry[]
  - libraryFilters: LibraryFilters
  - isLoadingLibrary: boolean

  # UI State
  - expandedCards: string[]
  - activeHerbDetail: string | null
  - addSupplementModalOpen: boolean
  - configureSupplementData: ConfigureSupplementData | null
  - scrollPosition: number

  # Chat
```

```
  - messages: SupplementMessage[]
  - isTyping: boolean
```

6.3 State Management Approach

Zustand Stores:

```
Stores:

1. scheduleStore
   - Today's schedule
   - Dose logging
   - Reminders
   - Adherence tracking

2. stackStore
   - Active supplements
   - Add/remove supplements
   - Effectiveness data
   - Synergy analysis
   - Safety status

3. recommendationsStore
   - Generated recommendations
   - Recommendation refresh
   - Add to stack flow

4. herbLibraryStore
   - Library entries
   - Search/filter
   - Herb detail views

5. supplementsUIStore
   - First visit flag
   - Expanded cards
   - Modal states
   - Messages
```

7. Component Requirements

7.1 Schedule Components

TodayScheduleCard

Displays daily supplement schedule:

- Date header

- Time windows (morning/afternoon/evening/bedtime)
- Doses within each window
- Status indicators
- Progress tracker
- Daily tip

States:

- Collapsed: Summary with upcoming dose
- Expanded: Full schedule with actions

ScheduledDoseItem

Individual dose in schedule:

- Supplement name
- Dosage amount
- Status indicator (✓, ○, ⏰)
- Time taken/scheduled
- Purpose (brief)
- Instructions
- Action buttons: [Log now] [Skip] [Remind me]

DoseStatusIndicator

- ✓ Taken: Green checkmark
- ○ Pending: Empty circle
- ⏰ Upcoming: Clock (within 1 hour)
- ✗ Missed: Red X (past scheduled time)

AdherenceWidget

- Daily progress: "2 of 3 taken"
- Weekly adherence: "85%"
- Visual progress bar

7.2 Stack Components

SupplementStackCard

List of active supplements:

- Count of active supplements
- List of supplement summaries
- Tap to see details

SupplementSummaryItem

- Herb icon/image
- Name
- Purpose (brief)
- Duration taking
- Effectiveness rating (stars)
- Tap to expand

SupplementDetailCard

Comprehensive supplement view:

- Header with image, name, Sanskrit
- Why taking (personalized)
- Ayurvedic properties
- Traditional + modern uses
- Dosage and timing
- Effectiveness tracking
- Safety information
- Quality guidance
- Action buttons

7.3 Recommendation Components

RecommendationsCard

Personalized suggestions:

- Context (what recommendations are based on)
- Priority recommendations list
- Expand to see all

RecommendationItem

- Herb icon/image
- Name
- Priority badge (High/Medium/Low)
- Brief purpose
- Brief "why for you"
- Action buttons: [Learn more] [Add to stack]

RecommendationDetailCard

Full recommendation reasoning:

- Profile match analysis
- What herb does
- Synergy with current stack
- Recommended dosage
- Expected timeline
- Actions: [Add to Stack] [Shop] [Not Interested]

7.4 Synergy Components

SynergyAnalysisCard

Stack synergy overview:

- Visual representation (Venn diagram or network)
- Overall synergy score
- Pair analyses
- Stack themes
- Gaps/suggestions

SynergyScoreBadge

- Score: 0-100
- Color-coded (excellent green, good teal, caution gold)
- Label: "Excellent combination!"

PairAnalysisItem

- Herb 1 + Herb 2
- Relationship: Complementary/Neutral/Competing
- Brief explanation
- Net benefit

7.5 Effectiveness Components

EffectivenessCard

Track supplement results:

- Per-supplement effectiveness
- Before/after charts
- Metric changes with ratings
- Continue/discontinue recommendation

EffectivenessChart

- Before/after comparison
- Line chart showing trend over time
- Metric value labels

MetricChangeItem

- Metric name
- Before → After values
- Change percentage
- Star rating
- Interpretation

EffectivenessRating

- Star display (1-5)
- Category label (Excellent, Good, Too Early, etc.)
- Color-coded

7.6 Safety Components

SafetyCheckCard

Stack safety analysis:

- Overall safety status
- Herb-herb interactions
- Herb-drug interactions (if medications provided)
- Contraindications
- Constitutional concerns
- Monitoring recommendations

SafetyStatusBadge

- ✓ Safe (green)
- ⚠️ Caution (gold)
- ⛔ Contraindicated (red)

InteractionItem

- Items involved
- Severity badge
- Description
- Recommendation

MedicationInputPrompt

- "If you're taking medications..."
- [Add medications to check interactions]

7.7 Herb Library Components

HerbLibraryCard

Browse herb database:

- Search input
- Filter chips (dosha, category)
- Results list

HerbListItem

- Herb image/icon
- Name (common and Sanskrit)
- Categories
- Dosha effect badges
- Status badges (in stack, recommended)

HerbDetailModal

Full herb information:

- Image
- Names and meanings
- Properties
- Uses
- Dosage
- Safety
- Shop link

7.8 Configuration Components

AddSupplementModal

Configure new supplement:

- Dosage selector
- Frequency selector
- Timing checkboxes
- Reminder setup
- Start date
- [Add to Stack] button

EditSupplementModal

Modify existing supplement:

- All configuration options

- [Save Changes] [Remove from Stack]

7.9 Animation Requirements

Animation	Duration	Easing	Trigger
Dose logged	300ms	ease-out	Log button tap
Checkmark appear	200ms	ease-out	Status change
Card expand	400ms	ease-out	Tap expand
Content fade-in	300ms	ease-in	After expand
Star rating fill	400ms	ease-out	Rating display
Chart draw	600ms	ease-out	Chart appears
Modal slide up	300ms	ease-out	Open modal
Badge pulse	1000ms	ease-in-out	New recommendation

8. API Requirements

8.1 REST Endpoints

Get Today's Schedule:

```
GET /api/v1/supplements/schedule

Response:
{
  "date": "2025-01-31",
  "doses": [
    {
      "supplementId": "supp_001",
      "supplementName": "Ashwagandha",
      "dosage": { "amount": 500, "unit": "mg", "form":
"capsule" },
      "timeWindow": "morning",
      "specificTime": "07:00",
      "status": "taken",
      "takenAt": "2025-01-31T07:15:00Z",
      "instructions": "With warm water, empty stomach",
      "anupana": "warm water"
    },
    ...
  ],
  "progress": 1,
  "total": 3,
```

```
  "adherenceRate": 85,
  "nextDose": {
    "supplementName": "Triphala",
    "timeUntil": "30 minutes"
  },
  "dailyTip": "Take Triphala at least 2 hours after
dinner..."
}
```

Log a Dose:

```
POST /api/v1/supplements/log

Request:
{
  "supplementId": "supp_001",
  "timestamp": "2025-01-31T07:15:00Z",
  "dose": { "amount": 500, "unit": "mg" },
  "notes": null
}

Response:
{
  "success": true,
  "doseLog": { ... },
  "updatedSchedule": { ... },
  "milestone": "30 days of Ashwagandha!" // if applicable
}
```

Get Supplement Stack:

```
GET /api/v1/supplements/stack

Response:
{
  "supplements": [
    {
      "id": "supp_001",
      "herbId": "herb_ashwagandha",
      "name": "Ashwagandha",
      "sanskritName": "Ashwagandha",
      "dosage": { ... },
      "schedule": [ ... ],
      "startDate": "2024-12-20",
      "daysTaking": 42,
      "reasonAdded": "Elevated Vata + stress patterns",
      "effectiveness": {
        "rating": 4,
```

```
        "category": "good"
      }
    },
    ...
  ],
  "synergyScore": 95,
  "safetyStatus": "safe"
}
```

Get Supplement Details:

```
GET /api/v1/supplements/:supplementId

Response:
{
  "supplement": { ... },
  "herb": { ... }, // full herb database entry
  "whyTaking": { ... }, // personalized reasons
  "effectiveness": { ... },
  "safetyInfo": { ... }
}
```

Get Recommendations:

```
GET /api/v1/supplements/recommendations

Response:
{
  "recommendations": [
    {
      "id": "rec_001",
      "herbId": "herb_brahmi",
      "herbName": "Brahmi",
      "sanskritName": "Brahmi",
      "priority": "high",
      "forImbalance": "elevated Pitta",
      "forGoal": "mental clarity",
      "reasoning": "Perfect for your Pitta constitution...",
      "benefits": [ ... ],
      "synergyWithStack": "Works well with Ashwagandha...",
      "suggestedDosage": { ... },
      "expectedTimeline": "2-4 weeks for initial effects"
    },
    ...
  ],
  "basedOn": {
    "prakriti": { ... },
    "vikriti": { ... },
```

```
    "goals": [ ... ],
    "season": "winter"
  }
}
```

Add Supplement to Stack:

```
POST /api/v1/supplements/add-to-stack

Request:
{
  "herbId": "herb_brahmi",
  "dosage": { "amount": 500, "unit": "mg", "form":
"capsule" },
  "schedule": [
    { "timeOfDay": "morning", "specificTime": "08:00",
"anupana": "ghee" },
    { "timeOfDay": "evening", "specificTime": "21:00",
"anupana": "warm milk" }
  ],
  "reminderEnabled": true,
  "startDate": "2025-02-01"
}

Response:
{
  "success": true,
  "supplement": { ... },
  "updatedStack": { ... },
  "updatedSafetyStatus": { ... }
}
```

Remove Supplement:

```
DELETE /api/v1/supplements/:supplementId

Request:
{
  "reason": "completed_course" | "not_effective" |
"side_effects" | "other",
  "notes": "..."
}

Response:
{
  "success": true,
  "removedSupplement": { ... },
  "updatedStack": { ... }
```

```
}
```

Get Synergy Analysis:

```
GET /api/v1/supplements/synergy

Response:
{
  "overallScore": 95,
  "pairAnalyses": [
    {
      "herb1": "Ashwagandha",
      "herb2": "Triphala",
      "synergy": "complementary",
      "explanation": "Ashwagandha builds, Triphala
cleanses...",
      "netBenefit": "Balance of building and detoxifying"
    },
    ...
  ],
  "stackThemes": ["Stress resilience", "Digestive health",
"Vitality building"],
  "gaps": ["Consider adding cooling herb for Pitta balance"],
  "optimizationSuggestions": [ ... ]
}
```

Get Effectiveness Data:

```
GET /api/v1/supplements/effectiveness/:supplementId

Response:
{
  "supplementId": "supp_001",
  "supplementName": "Ashwagandha",
  "daysTaking": 42,
  "adherenceRate": 95,
  "rating": {
    "overall": 4,
    "confidence": "high"
  },
  "category": "good",
  "metricChanges": [
    {
      "metricName": "HRV",
      "before": 50,
      "after": 62,
      "changePercent": 24,
      "rating": 5,
```

```
      "interpretation": "Significant improvement in stress
resilience"
    },
    ...
  ],
  "subjectiveChanges": [ ... ],
  "analysis": "Ashwagandha is working well for you...",
  "recommendation": "continue",
  "recommendationReason": "Continue for 90 days total for
deep nourishment"
}
```

Get Safety Check:

```
GET /api/v1/supplements/safety-check

Response:
{
  "overall": "safe",
  "herbHerbInteractions": [],
  "herbDrugInteractions": [],
  "contraindications": [],
  "constitutionalConcerns": [
    {
      "concern": "Pitta elevation",
      "explanation": "You have 60% Pitta baseline.
Ashwagandha is warming.",
      "recommendation": "Monitor for increased heat, acidity,
irritability"
    }
  ],
  "monitoringRecommendations": [
    "Watch for signs of increased Pitta"
  ]
}
```

Get Herb Library:

```
GET /api/v1/supplements/library?dosha=vata&category=adaptogen

Response:
{
  "herbs": [
    {
      "id": "herb_ashwagandha",
      "commonName": "Ashwagandha",
      "sanskritName": "Ashwagandha",
      "imageUrl": "...",
```

```
      "categories": ["adaptogen", "rasayana", "nervine"],
      "doshaEffects": {
        "vata": "decreases",
        "pitta": "increases",
        "kapha": "decreases"
      },
      "primaryUses": ["Stress", "Sleep", "Vitality"],
      "briefDescription": "Premier adaptogen for stress and
Vata...",
      "isInUserStack": true,
      "isRecommended": false
    },
    ...
  ],
  "totalCount": 24
}
```

8.2 WebSocket Events (Optional)

```
WebSocket Events:

dose.reminder
{
  "supplementId": "supp_001",
  "supplementName": "Triphala",
  "scheduledTime": "21:00",
  "message": "Time to take Triphala"
}

dose.logged
{
  "supplementId": "supp_001",
  "timestamp": "2025-01-31T21:05:00Z",
  "progress": { "taken": 2, "total": 3 }
}

effectiveness.updated
{
  "supplementId": "supp_001",
  "newRating": 4,
  "milestone": "6 weeks complete"
}
```

9. Success Metrics

9.1 Engagement Metrics

Metric	Target	Measurement
Section visits per week	> 5	Users visiting Supplements weekly
Active supplement users	> 40%	Users with at least 1 active supplement
Dose logging rate	> 70%	Logged doses / scheduled doses
Schedule view frequency	Daily	Users checking schedule daily
Return rate	> 60%	Users returning within 7 days

9.2 Adherence Metrics

Metric	Target	Measurement
Weekly adherence	> 80%	Average across all users
Perfect adherence days	> 50%	Days with 100% compliance
Missed dose rate	< 15%	Doses missed (not logged or skipped)
Reminder effectiveness	> 60%	Doses logged within 30min of reminder

9.3 Feature Usage Metrics

Metric	Target	Measurement
Recommendation views	> 50%	Users who view recommendations
Recommendation acceptance	> 20%	Recommendations added to stack
Effectiveness tracking views	> 40%	Users who check effectiveness
Safety check views	> 30%	Users who view safety analysis
Herb library usage	> 25%	Users who browse library

9.4 Action Metrics

Metric	Target	Measurement
Supplements added	> 1.5/ user	Average supplements in stack
Supplements completed	Track	Supplements taken for full duration
Shop click-through	> 15%	Users clicking shop links
Supplement continuation	> 60%	Supplements continued past 30 days

9.5 Technical Metrics

Metric	Target	Measurement
Schedule load time	< 1s	Time to display today's schedule
Dose logging time	< 500ms	Time to confirm logged dose
Recommendation generation	< 2s	Time to generate recommendations
Safety check time	< 1s	Time to complete safety analysis
Error rate	< 1%	Failed operations

10. Implementation Considerations

10.1 Notification System

Dose Reminders:

- Configurable per supplement
- Default: 15 minutes before scheduled time
- Follow-up if not logged within 30 minutes
- Respect device notification settings

Milestone Notifications:

- "30 days of Ashwagandha!"
- "Time for effectiveness check-in"
- "Refill reminder: Triphala running low"

Implementation:

- Use device push notifications
- Schedule locally for reliability
- Sync with server for consistency
- Allow snoozing and dismissal

10.2 Effectiveness Tracking Integration

Data Sources:

- Health metrics from wearables (HRV, sleep)
- Self-reported symptoms
- Dosha assessments over time
- Agni and Ojas scores

Correlation Approach:

- Track metrics from supplement start date
- Compare to baseline (30 days before)
- Account for adherence in analysis
- Consider confounding factors (other supplements, season)

Minimum Data Requirements:

- At least 14 days of supplement use
- At least 70% adherence
- Relevant metrics available

10.3 Safety Database

Interaction Database:

- Comprehensive herb-herb interactions
- Herb-drug interactions (common medications)
- Regular updates from authoritative sources
- Conservative approach (flag uncertain interactions)

Sources:

- Natural Medicines Database
- Ayurvedic pharmacopeia
- Published research
- Traditional contraindications

10.4 Offline Handling

When Offline:

- Schedule viewable from cache
- Dose logging works (queues for sync)
- Cached herb information available
- Recommendations may be stale (show warning)

When Back Online:

- Sync queued dose logs
- Update schedule from server
- Refresh recommendations if stale
- Update safety status

10.5 Performance Optimization

Data Loading:

- Load today's schedule first (critical path)
- Lazy-load stack details
- Cache herb database locally
- Prefetch likely herb details

Image Handling:

- Lazy load herb images
- Use thumbnails in lists
- Full images in detail views
- Offline cache for active supplements

10.6 Accessibility Requirements

Screen Reader:

- Schedule times and status announced
- Effectiveness ratings described
- Safety warnings prominent
- Actions clearly labeled

Motor Accessibility:

- Large tap targets for dose logging
- Swipe actions for quick log
- No time-sensitive interactions
- Voice command support (future)

Visual Accessibility:

- WCAG AA contrast
- Color + icon for status (not color alone)
- Support Dynamic Type
- High contrast mode support

10.7 Analytics Events

Event	Properties	Purpose
`supplements_opened`	source, is_first_visit	Track entry
`dose_logged`	supplement_id, on_time	Adherence tracking
`dose_skipped`	supplement_id, reason	Skip patterns
`schedule_viewed`	time_of_day	Engagement
`supplement_detail_viewed`	supplement_id	Interest
`recommendation_viewed`	herb_id, priority	Recommendation engagement
`recommendation_added`	herb_id	Conversion
`recommendation_dismissed`	herb_id, reason	Preference learning
`effectiveness_viewed`	supplement_id	Engagement
`safety_viewed`	—	Safety awareness
`library_searched`	query, filters	Discovery
`shop_clicked`	herb_id, product_id	Commerce
`supplement_removed`	supplement_id, reason	Churn analysis

10.8 Testing Priorities

Unit Tests:

- Dose logging logic
- Adherence calculation
- Synergy scoring
- Safety checking logic

Integration Tests:

- Full logging flow
- Add supplement to stack flow
- Recommendation to stack conversion
- Effectiveness data aggregation

E2E Tests:

- First-time user education flow
- Complete daily supplement routine
- Add supplement from recommendation
- Safety check with medications

Manual QA:

- Notification delivery and timing
- Effectiveness chart accuracy
- Safety warning appropriateness
- Herb information accuracy
- Recommendation relevance

Appendix A: Common Ayurvedic Herbs Reference

Adaptogens

Herb	Sanskrit	Best For	Caution For
Ashwagandha	अश्वगन्धा	Vata, stress	High Pitta
Shatavari	शतावरी	Pitta, women's health	Kapha, congestion
Tulsi	तुलसी	Kapha, immunity	—
Brahmi	ब्राह्मी	Pitta, mental clarity	Kapha

Digestives

Herb	Sanskrit	Best For	Caution For
Triphala	त्रिफला	All doshas, detox	—
Trikatu	त्रिकटु	Kapha, Agni	Pitta
Ginger	अदरक	Vata, Kapha	High Pitta
Fennel	शतपुष्पा	Pitta, digestion	—

Nervines

Herb	Sanskrit	Best For	Caution For
Brahmi	ब्राह्मी	Pitta, Vata	Kapha excess
Jatamansi	जटामांसी	Vata, sleep	—
Shankhapushpi	शंखपुष्पी	All, memory	—

Rejuvenatives (Rasayanas)

Herb	Sanskrit	Best For	Caution For
Ashwagandha	अश्वगन्धा	Vata, Ojas	High Pitta
Amalaki	आमलकी	Pitta, immunity	—
Guduchi	गुडूची	All, immunity	—
Chyawanprash	च्यवनप्राश	All, vitality	Diabetes (sugar content)

Appendix B: Timing Reference

Best Times for Different Herbs

Timing	Sanskrit	Best For
Empty stomach	Abhakta	Rasayanas, adaptogens, detox
Before meals	Pragbhakta	Digestive stimulants
With meals	Saabhakta	Minerals, nutrients
After meals	Adhobhakta	Digestive aids
Between meals	Antarabhakta	Therapeutic herbs
At bedtime	Nishi	Nervines, laxatives

Carriers (Anupanas) by Purpose

Carrier	Best For	Enhances
Warm water	General	Absorption
Warm milk	Building	Ojas, nourishment
Ghee	Deep tissue	Brain, joints
Honey	Reducing	Kapha, quick action
Aloe vera	Cooling	Pitta
Coconut water	Cooling	Pitta, hydration

Appendix C: Effectiveness Rating Criteria

Rating Scale

Rating	Category	Criteria
⭐⭐⭐⭐⭐	Excellent	Clear, measurable improvements in target metrics (>20% change)
⭐⭐⭐⭐	Good	Noticeable improvements (10-20% change)
⭐⭐⭐	Too Early	Less than minimum effective period (usually <4 weeks)
⭐⭐	Minimal	Limited benefits (<10% change) after adequate time
⭐	Not Effective	No measurable benefits after adequate time

Minimum Effective Periods

Herb Type	Minimum Period
Adaptogens	4-6 weeks
Digestives	2-4 weeks
Nervines	2-4 weeks
Rasayanas	6-8 weeks
Detox herbs	4-6 weeks

Appendix D: Safety Severity Levels

Interaction Severity

Level	Description	Action
Mild	Minimal clinical significance	Inform user, no change needed
Moderate	May require monitoring	Warn user, suggest monitoring
Severe	Potentially dangerous	Strongly advise against, require confirmation

Contraindication Severity

Level	Description	Action
Caution	Use with care, monitor	Yellow warning, education
Avoid	Should not use	Red warning, block or require override

End of Document